

ORBIT — OFFLINE MULTIMODAL RAG ASSISTANT

COMPLETE USER MANUAL, INSTALLATION GUIDE & APP WALKTHROUGH

1. Executive Overview & Mission

Orbit is an enterprise-grade, local-first Retrieval-Augmented Generation (RAG) assistant built for complete data privacy. Powered by dense vector embeddings (*nomic-embed-text*), sparse keyword reranking (*Rank-BM25*), and local LLMs (*llama3.2:3b*), Orbit analyzes your documents, spreadsheets, images, audio, video, and SQL databases without transferring any information to cloud servers.

Platform Name	Orbit (Offline Multimodal RAG Assistant)
Live Web Portal	https://offlinerag-psi.vercel.app
Direct Download URL	https://github.com/kalakonda-akshay/RAG/releases/latest/download/OfflineRAGAssistant_Setup.zip
Supported Formats	29 File Extensions (PDF, DOCX, XLSX, PPTX, MP3, MP4, PNG, SQLite)
Local AI Engine	Ollama (llama3.2:3b + nomic-embed-text 768-dim)

2. Download & Installation Guide

■ **Air-Gapped Privacy Guarantee:** Orbit runs 100% locally on your computer CPU/GPU. No internet connection is required after initial model download.

Step 01: Download Setup Package

Download *OfflineRAGAssistant_Setup.zip* (1.9 GB) from <https://offlinerag-psi.vercel.app> or GitHub Releases.

Step 02: Extract & Environment Setup

Extract the zip file to your folder (e.g. *C:\OrbitAssistant*) and run the automated setup wizard in terminal:
`python launcher/first_run_setup.py`

Step 03: Pull Local Ollama LLM Models

Run the following terminal commands to pull local models:
`ollama pull llama3.2:3b`
`ollama pull nomic-embed-text`

Step 04: Launch Web Interface

Start the application server: `python launcher/run_app.py`
Access URL: **`http://localhost:8501`**

3. Opening the App — Layout & Interface Walkthrough

When you launch Orbit at `http://localhost:8501`, you enter a modern 3-column dark-mode workspace. Here is a walkthrough of what you see after opening the app:

Workspace Component	Description & Features Available After Opening
Left Knowledge Rail (Workspace & File Manager)	• Brand Header: Orbit logo and version indicator. • Workspace Switcher: Select between <i>documents</i>, <i>research</i>, <i>finance</i>, and <i>engineering</i>. • Indexed File Cards: Displays loaded files with type icons (PDF, XLSX, DOCX). • Upload Dropzone: Drag & drop any of the 29 supported file types. • AI Personas: Switch between <i>Technical Analyst</i>, <i>Tutor</i>, and <i>Legal Officer</i>. • Export Button: 1-Click PowerPoint deck export (<i>Executive_Deck.pptx</i>).
Center Q&A; Thread (Feature Dashboard & Chat)	• Scope Banner: Shows active scope (e.g. <i>scoped to research (1 file)</i>). • 12 Feature Tabs: Quick switcher for Chat, Voice Call, Web Archiver, Mind Map, Topics, Translate, Battle, Quiz, SQL DB, PII, Compare, Reports. • Message Stream: Displays grounded AI answers with inline [1], [2] citation chips. • Interactive Source Pills: Clickable source buttons (e.g. [1] <i>Company Financials Q1 (p.4)</i>). • Prompt Input Bar: Type grounded questions over documents.
Right Grounding Rail (Source Inspection)	• Grounding Gauge: Shows citation verification rating (e.g. <i>0.91 high</i>). • Relevance Cards: Shows retrieved text passages with cosine similarity scores. • Highlighted Text: Displays exact matched text with <code><mark></code> highlights. • Page & Chunk Markers: Shows exact page numbers and chunk boundaries.

4. Live Interface & Screenshot Tour

Figure 1: Grounded Q&A; Thread & Interactive Source Pills

Displays the main chat thread, inline citation badges **[1]**, **[2]**, interactive document pills, and prompt input box:

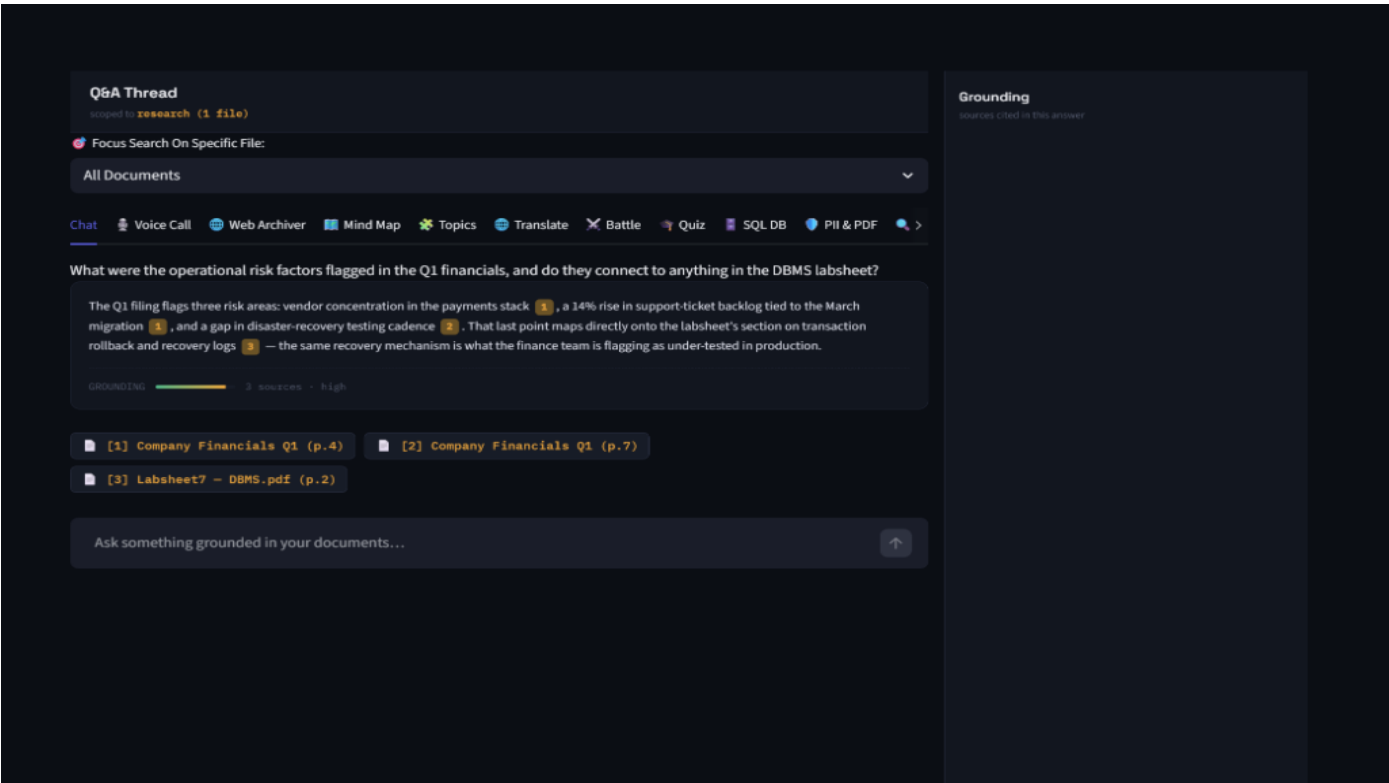


Figure 2: Real-Time Voice Call Mode

Displays the hands-free microphone audio recording interface powered by local Whisper speech-to-text:

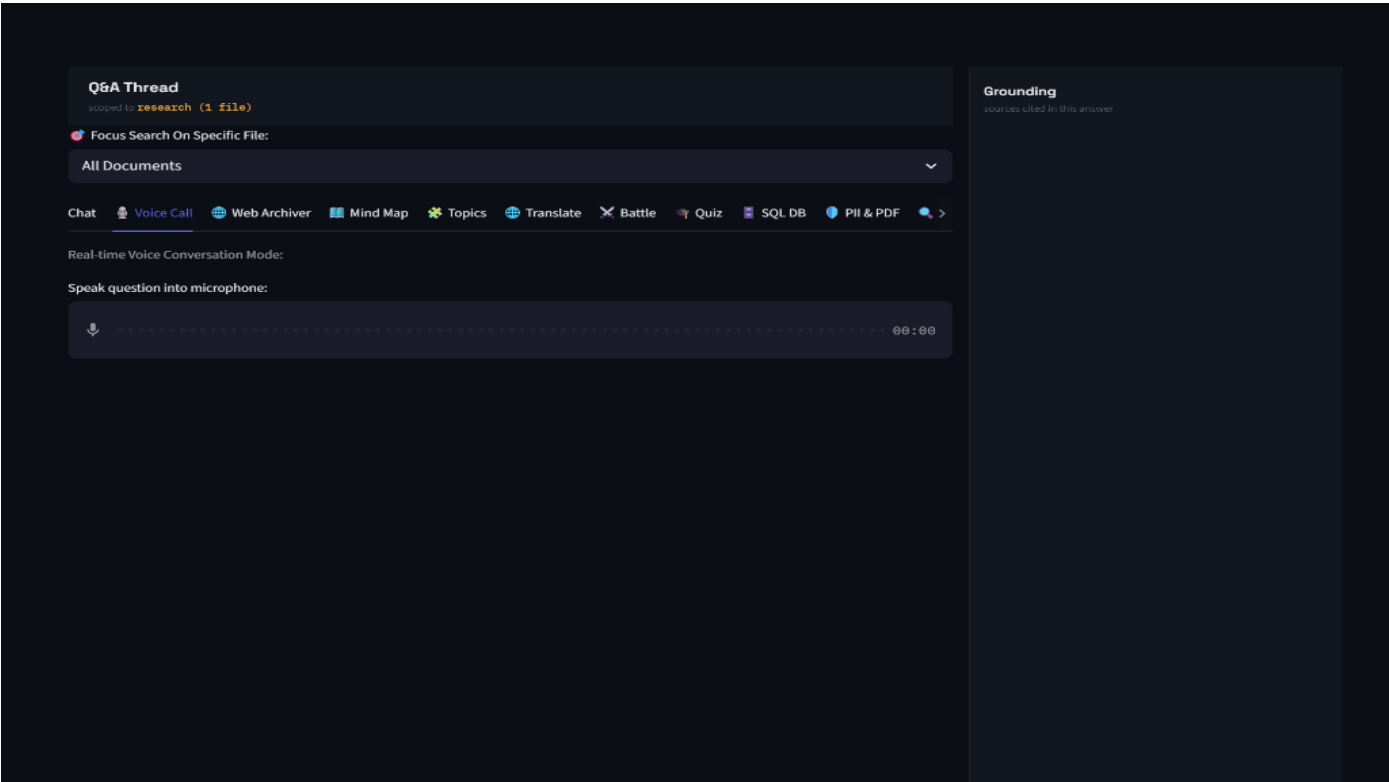
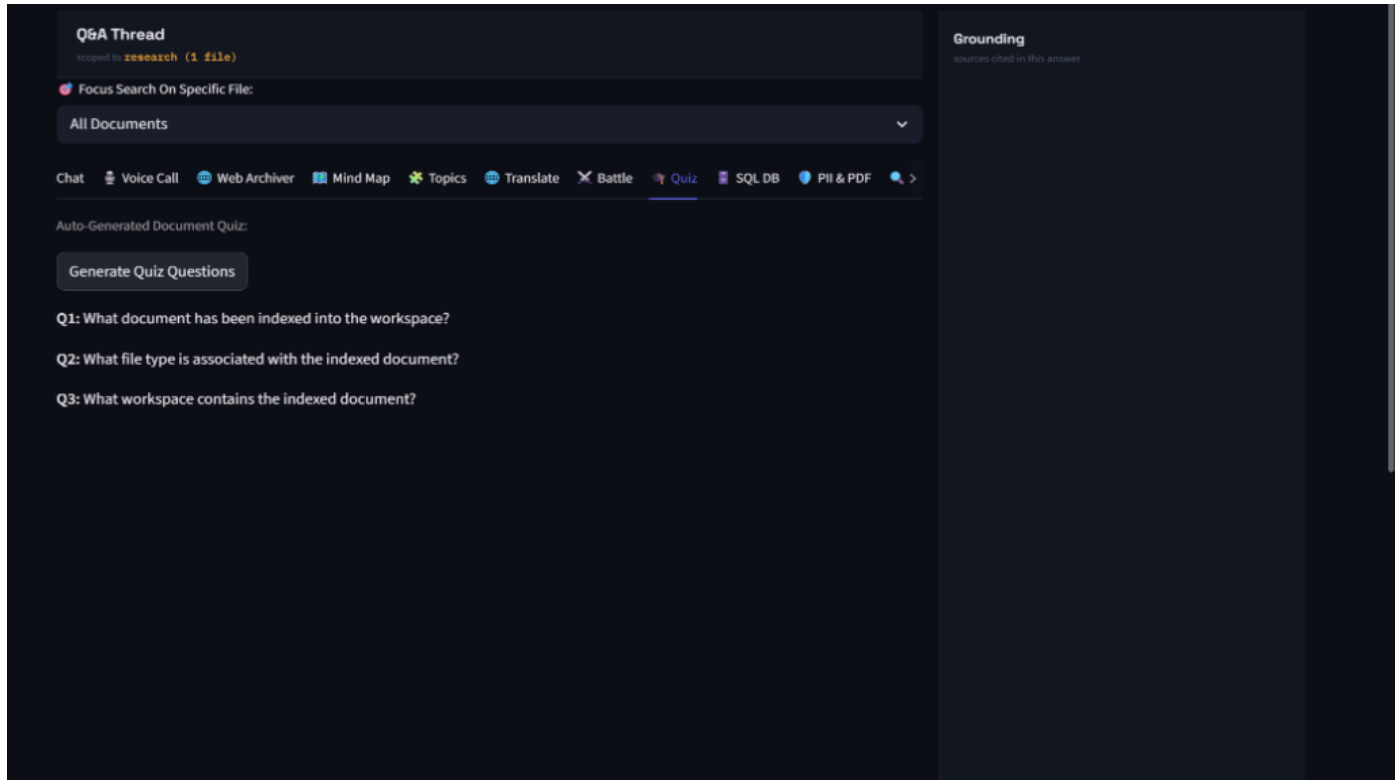


Figure 3: Auto-Generated Document Quiz Workspace

Displays practice test questions auto-generated from indexed document contents:



5. Complete Feature Reference (12 Feature Tabs)

- **Grounded Q&A; Chat:** Dense vector search + Rank-BM25 hybrid retrieval + Self-RAG hallucination verification.
- ■ **Voice Call Mode:** Real-time microphone recording and local Whisper speech transcription.
- **Web Archiver:** Parses and indexes offline web pages and documentation HTML.
- ■ **Mind Map Generator:** Generates visual Mermaid.js mind maps from document concepts.
- **Topic Cluster Discovery:** Performs K-Means semantic clustering across document text.
- **Multilingual Translator:** Translates document passages into 10+ target languages offline.
- ■ **Model Battle Mode:** Compares responses side-by-side across multiple local LLMs.
- **Auto Document Quiz:** Generates practice test questions with answer keys.
- ■ **Natural Language SQL Engine:** Queries SQLite (.db/.sqlite) tables using auto-generated SQL.
- ■ **PII Redactor & PDF Exporter:** Redacts SSNs, credit cards, emails, and exports sanitized PDFs.
- **Document Comparator:** Side-by-side comparison of two documents highlighting diffs.
- **Report & PPTX Exporter:** 1-click PowerPoint presentations (.pptx) and executive brief summaries.

6. Supported File Formats (29 Extensions)

- **Documents & Presentations (5):** .pdf, .docx, .pptx, .ppt, .md
- **Spreadsheets & Tables (3):** .xlsx, .xls, .csv
- **Images & OCR (7):** .png, .jpg, .jpeg, .webp, .bmp, .tiff, .gif
- **Audio Recordings (3):** .wav, .mp3, .m4a
- **Video Files (4):** .mp4, .mkv, .mov, .avi
- **Code & Text Markup (7):** .txt, .py, .js, .json, .html, .xml, .sql
- **Relational DBs (2):** .sqlite, .db

7. Algorithmic Architecture & RAG Pipeline

1. **Multimodal Ingestion:** Extracts text, tables, image OCR, and audio voice files.
2. **Recursive Chunker:** Overlapping sliding window (500 tokens / 50 overlap).
3. **Dense Embeddings:** nomic-embed-text 768-dimensional dense vectors.
4. **Hybrid RRF Search:** Dense Cosine Similarity + Sparse Rank-BM25 Reciprocal Rank Fusion.
5. **Agentic Multi-Hop RAG:** Recursive sub-query splitting.
6. **Local LLM Generator & Self-RAG Critic:** Ollama llama3.2:3b fact audit.
7. **GraphRAG:** Entity-relationship triple extraction (Source -> Relation -> Target).